

KEY DEVELOPMENT

STEAM, IRONCLADS, AND THE FIRST BATTLESHIPS

In the century between the Napoleonic Wars and the outbreak of World War I, new technology transformed naval warfare. Sail gave way to steam, wooden hulls were superseded by metal ones, and high-explosive shells replaced cannonballs.

The sailing ship navies of Nelson's era (see p.196) did not disappear overnight; at first, steamships were only viable in coastal and inland waters. Their large paddle wheels limited the space available for gunports and were vulnerable to enemy fire, but steam ships had the clear advantage of being able to manoeuvre at will, even in a dead calm. The introduction, in the 1840s, of screw propellers as an alternative to paddle wheels led to the adoption of steam engines as a standard feature of warships, although initially as an auxiliary to the sails.

RISE OF THE IRONCLAD

The 1840s also saw the Paixhans gun revolutionize naval armament: it was the first naval gun designed to fire explosive shells instead of solid shot. In response, during the 1850s, some wooden-hulled ships were armoured with thick iron plate above the waterline. These ironclad ships were deemed a

success during the Crimean War (1853–56), fought between Russia and the allied forces of Britain, France, the Ottoman Empire, and Kingdom of Sardinia. In 1859, France built the first ironclad battleship, *La Gloire*, which was matched by the British HMS *Warrior* the following year.

The American Civil War began in 1861. The Confederates rebuilt an existing ship as the ironclad CSS *Virginia*, while their opponents sponsored a radical new design, the USS *Monitor* – a metal raft equipped with two Dahlgren guns in a rotating turret. At the Battle of Hampton Roads, in March 1862, CSS *Virginia* sank two conventional Union frigates with frightening ease, but then fought a stalemated duel with USS *Monitor* in what was to be the first battle between two ironclad steamships.

Experiments with ironclad ships continued throughout the Civil War: the Union forces built more monitors (named after the original), as well

KEY EVENTS

1800–1950

■ **1841** An iron-hulled steam-powered gunboat, *Nemesis*, plays a key role in the British defeat of China in the First Opium War.

■ **1853** Making the first use of explosive shells in a naval battle, a Russian squadron destroys 11 Turkish ships at the Battle of Sinope, during the Crimean War.

■ **1897** The US Navy adopts the first successful powered submarine, designed by Irish-American John Holland.

■ **1904** Japanese destroyers armed with Whitehead torpedoes make a surprise attack on the Russian fleet at Port Arthur.

■ **1911** Britain's Royal Navy decides that its new *Queen Elizabeth*-class battleships will use oil as fuel, instead of coal.

▼ METAL AGAINST METAL

In 1864, during the American Civil War, Union admiral David Farragut penetrated Confederate-held Mobile Bay with a squadron of metal monitors and wooden ships. One monitor was sunk by a mine, but Farragut disabled the Confederate ironclad CSS *Tennessee*.

